# SimpliSpeak: User-Controlled Text Simplification for Audio-First Accessible Reading

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## Abstract

Screen readers make digital text audible, but they do not make complex language easier to understand. We present SimpliSpeak, an audio-first reading application in which blind and low-vision listeners decide, while listening, how simple a text should be. Building on a Phase 1 low-fidelity Figma prototype, the Phase 2 implementation is a web application that runs in a phone browser. Text can be pasted, typed or photographed. A large language model (Google Gemini) produces three simplified levels—Light, Simple and Very Simple—in English or Hindi, and the original wording always remains available. Text is read one sentence at a time; listeners can change the level without losing their position, ask for word explanations and return to the exact original. A rule-based protected-facts guard checks that numbers, dates, amounts and negations survive simplification and, when they do not, warns the listener and falls back to a fuller level. The application can be operated through English and Hindi voice commands, screen-reader-style touch gestures, large buttons or the user's own screen reader, and it includes an instrumented study mode. Behaviour was verified through automated tests and team testing on a real phone. No study with blind or low-vision participants has yet been conducted, so no claims about comprehension, workload or usability are made.

## CCS Concepts

• **Human-centered computing** → **Accessibility**.

## Keywords

Accessibility, blind and low-vision users, text simplification, screen readers, audio-first interfaces, voice user interfaces, Hindi, large language models

## 1 Introduction

Screen readers give blind and low-vision people access to digital text by converting it into speech. Access to the words, however, does not guarantee access to their meaning. When a document contains long sentences, unfamiliar terminology or dense instructions, the same difficulty remains in the audio. This is especially relevant for medical leaflets, bank letters, legal notices and administrative documents, where people must understand and act on specific information.

Listening is also sequential: spoken words disappear as they are heard, and it is hard to skim ahead or revisit an earlier phrase. Screen-reader users adopt non-linear browsing strategies rather than consuming content from beginning to end [4], and their listening speeds can differ considerably from typical speech rates [2]. An accessible reading interface should therefore support both efficient navigation and comprehension.

SimpliSpeak takes an audio-first approach. Text is simplified in short portions, and the listener—not the system—chooses how simple it should become. The listener can ask for simpler wording, more detail, an explanation of a difficult word, or a repetition, and can always return to the exact original wording. This report describes the Phase 2 implementation of this concept as a working phone web application and separates what was tested on a real phone, what was verified only in automated emulator tests, and what remains future work.

The contributions of this work are:

(1) A listener-controlled simplification interaction with four levels (Original, Light, Simple, Very Simple), in which changing the level keeps the current reading position, and on-demand word explanations.
(2) A protected-facts guard that checks simplified text for lost numbers and negations and falls back to a fuller level with a spoken warning.
(3) Three ways of operating the application without relying on sight—touch exploration, large buttons and the user's own screen reader—combined with voice commands and earcons.
(4) A bilingual English and Hindi experience covering simplified text, application speech and voice commands.
(5) An instrumented study mode implementing a counterbalanced evaluation for future testing with blind and low-vision users.

## 2 Related Work

### 2.1 Text Simplification and Cognitive Accessibility

Research on text simplification covers lexical and syntactic techniques for reducing linguistic complexity while retaining important information [12]. Cognitive load theory motivates limiting the amount of information that must be held and processed at once [14]. WCAG 2.2 includes a reading-level success criterion [18], and W3C guidance on cognitive and learning disabilities stresses clear, understandable content [6]. SimpliSpeak applies these ideas by offering several user-selectable simplification levels while keeping the original text available.

### 2.2 Non-Visual and Auditory Interaction

Borodin et al. surveyed how screen-reader users browse the web and the navigation problems they face [4]; this informs SimpliSpeak's swipe-based exploration, in which each control announces its name and position before it is activated. Earcons, structured non-speech

sounds, can convey interface events without additional speech [3]; SimpliSpeak uses thirteen earcons for events such as level changes, microphone activation, safety warnings and the end of the text. Mobile screen readers such as TalkBack and VoiceOver define established gesture conventions [15, 17], which SimpliSpeak's touch mode imitates and its screen-reader mode defers to.

### 2.3 Existing Reading Tools

Several applications already support accessible reading (Table 1). They provide speech synthesis, text recognition, display customisation and, increasingly, AI summaries, but none lets the listener control the linguistic complexity of spoken content sentence by sentence. SimpliSpeak's contribution is therefore not any single feature but their integration into one audio-first reading interaction. Whether this integration offers measurable advantages would require a comparative user evaluation.

## 3 Conceptual Design

### 3.1 Target Users

The intended users are blind and low-vision people who use audio interfaces to access written information, with differing experience of assistive technology. Two illustrative personas guided the design. *Asha* is an experienced screen-reader user who operates her phone independently but struggles with dense bank letters and medicine leaflets. *Ravi* has recently lost his sight and is learning audio tools while managing health paperwork; he benefits from slower speech, repetition and simple ways to ask for help. The personas were derived from background reading and team discussion, not from interviews.

### 3.2 Design Goals

(1) **Understand without restarting:** short portions, repetition and backward navigation.
(2) **Control simplification:** several levels that the listener can choose and change at any time.
(3) **Trust transformed content:** exact source text always available and critical facts protected.
(4) **Resolve vocabulary:** explanations only when requested.
(5) **Interact without sight:** voice, gestures, keyboard input and concise audio cues.
(6) **Use the listener's language:** simplified text, speech and commands in Hindi as well as English.

### 3.3 Core Interaction

The primary interaction is a listen–adjust loop. The user brings text into the application, listens to one sentence at a time and adjusts when needed: simpler, more detail, explain a word, repeat or original. After the last sentence the application asks whether the text was clear; if the answer is "no", it restarts one level simpler.

### 3.4 Three Modes of Use

A web page cannot receive ordinary touch gestures while a phone screen reader is running, and a person who cannot see the screen cannot find a button by looking for it. The opening screen therefore offers three modes:

- **Touch mode** (default): swiping moves between controls and announces each one; double tap activates it; holding re-reads the screen and the available gestures.
- **Button mode**: large buttons for low-vision users and sighted helpers, with the application still speaking.
- **Screen-reader mode**: the application's own speech is turned off so that TalkBack, VoiceOver or another screen reader reads the labelled controls in its own voice and speed.

All three modes provide a voice-guided start and accept voice commands.

## 4 System Implementation

### 4.1 System Overview

SimpliSpeak has two parts (Figure 1). The *client* is a single-page web application written in plain HTML, CSS and JavaScript that runs in the phone's browser without installation; it handles input, interaction, the reading state and speech. The *server* is a lightweight Python service that performs the language processing. Each document passes through four stages:

(1) **Input:** the text is pasted, typed or photographed on the phone (Section 4.2).
(2) **Segmentation:** the server splits the text into sentences, each of which becomes one navigable part.
(3) **Simplification:** a single request to the Gemini API [7] returns all three levels and a glossary; answers are cached, and a rule-based English simplifier is used when the AI service is unavailable (Sections 4.3 and 4.5).
(4) **Verification:** the protected-facts guard checks every level and marks those that lose a fact (Section 4.4).

Because every level arrives together, level changes, navigation and word help then happen instantly on the phone without further requests.

### 4.2 Text Input and Preparation

Three input routes are implemented: pasting from the clipboard, typing into a text field, and photographing printed paper. For photos, the phone's camera application opens; the image is scaled to at most 1,800 pixels, sent as JPEG to the multimodal model and transcribed in reading order. If the AI service is unavailable, Tesseract.js performs OCR in the browser [13]. If fewer than four words are found, the user is asked to retry with better lighting and the whole page in view. The accessibility version also offers a live camera view with spoken guidance and automatic capture when the page is steady. Texts are limited to 3,000 words.

The server splits text into paragraphs and then sentences. Sentence boundaries are periods, exclamation marks, question marks and the Hindi danda (the full stop of Devanagari) when followed by a capital letter, digit or Devanagari letter; abbreviations such as "Dr.", "Mr.", "Rs." and "No." do not end a sentence. Each sentence becomes one navigable part and its original wording is stored unchanged.

### 4.3 Simplification Levels and Word Help

One request per document asks the model for three versions of every numbered sentence plus a glossary, returned in a fixed JSON structure. All levels are generated up front, so the listener can

**Table 1: Existing accessible-reading tools compared with SimpliSpeak.**

| Tool | Available capabilities | Difference in SimpliSpeak |
|---|---|---|
| NaturalReader [10] | Text-to-speech, OCR, adjustable speed, sentence repetition, highlighting, AI summaries, quizzes and document question answering. | Selectable simplification levels that change sentence by sentence without losing the reading position; voice and touch control. |
| Voice Dream Reader [16] | Document reading, adjustable speed, pronunciation control, highlighting, notes, offline listening and VoiceOver support. | AI-based simplification combined with contextual word explanations and voice-controlled reading. |
| Microsoft Immersive Reader [9] | Read aloud, line focus, picture dictionary, translation, syllabification and display customisation. | Listener control over the linguistic complexity of spoken content, with access to the original wording. |
| Seeing AI [11] | Camera-based text recognition, document scanning, reading and question answering. | Text extraction followed by adjustable simplification, fact checks and audio-first reading controls. |

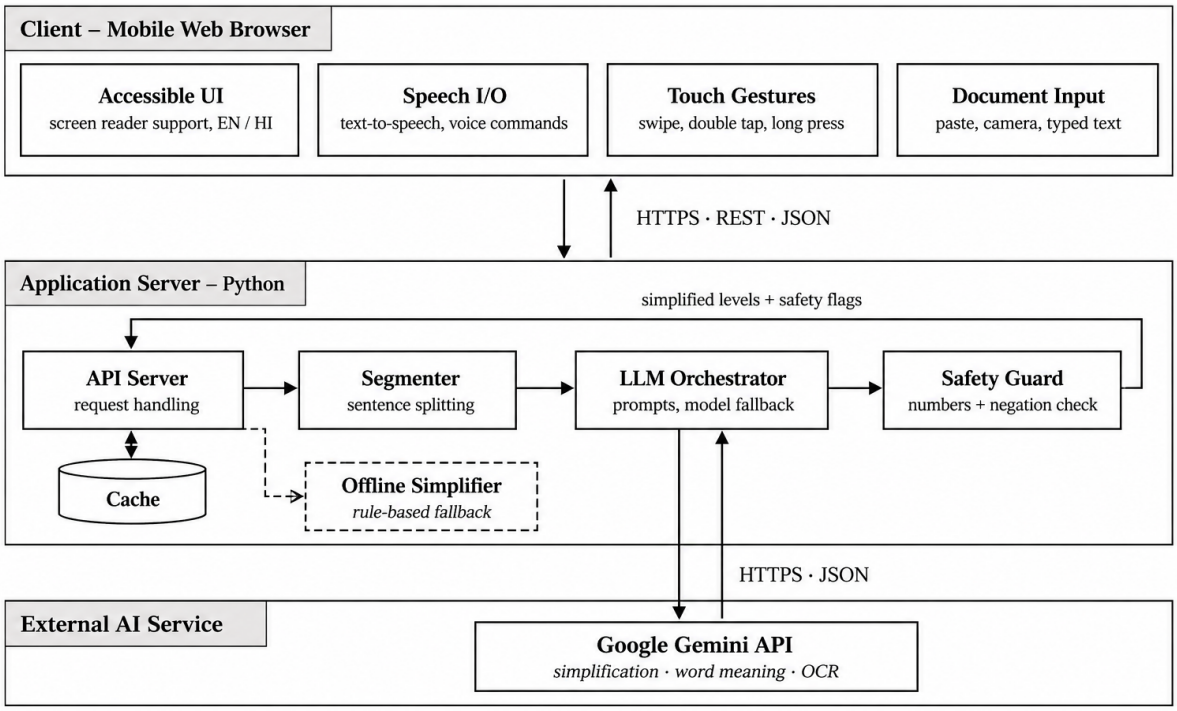



**Figure 1: Architecture of SimpliSpeak. The phone client handles input, interaction, reading state and speech. The server segments the text, obtains simplified levels and a glossary from Gemini models, checks every level with the protected-facts guard, caches answers and falls back to rule-based simplification.**

switch levels instantly without losing the current sentence. The default level is Simple. The four levels are:

- **Original:** the source wording, unchanged.
- **Light:** plain words, roughly the same structure.
- **Simple:** active voice, sentences of about twelve words or fewer.
- **Very Simple:** one idea per statement, about eight words; long sentences may be split.

The prompt instructs the model to preserve numbers, doses, dates, amounts and names, not to drop or soften warnings and deadlines, not to add advice, and to write for listening in a respectful adult tone. For Hindi it requests everyday spoken Hindi in Devanagari; glossary entries keep the English term beside the Hindi explanation. Word explanations are generated on request and limited to about twenty words in the selected language.

### 4.4 Protected-Facts Guard and Level Fallback

For every sentence and level, the guard extracts numeric facts from the original: digits (with separators removed), English number words ("one" to "twelve", "once", "twice", "half"), supported Hindi number words and Devanagari digits. If the original contains a negation or prohibition ("not", "never", "no", "cannot", "stop", "discontinue"), the simplified version must also contain a recognised English or Hindi negation.

A level that fails is marked *unsure* together with the facts it lost. When the selected level is unsure, the phone plays a soft double beep, speaks a warning, shows the lost facts on screen and reads the next fuller level that passed (Very Simple → Simple → Light); if none passes, the original is read. For example, in the Hindi electricity-bill example the Very Simple version omitted the billing period "1 to 30 September", so the fuller Hindi version was used. The guard detects omissions of supported facts only; it cannot establish semantic equivalence and does not detect changed names or added instructions.

### 4.5 Reliability

The server uses a configurable chain of Gemini models. A model that reports its daily limit is skipped until the limit resets; a busy model gets one quick retry and is then skipped temporarily; the chain gives up after about ninety seconds. Successful answers are cached under a hash of task, language and text, so repeated texts and examples need no new request, even after a restart. When no model responds, a rule-based English simplifier and word list take over, and a plain-language status message explains the limitation.

### 4.6 Speech Output and Voice Input

Speech uses the Web Speech API [19]. Because mobile browsers require a user gesture before speaking, the first tap initialises audio, and a fallback button appears if speech has not started within six seconds. Hindi uses an installed `hi-IN` voice; English uses the phone's default voice. Speed is adjustable from 0.6× to 1.8×. Thirteen earcons generated with the Web Audio API mark events: for example, a soft tick for a new part, rising or falling notes for simpler or more detail, a short beep when the microphone listens, a soft double beep for a safety warning and three rising notes at the end of the text.

Voice commands use speech recognition in Indian English (`en-IN`) or Hindi (`hi-IN`). Phrases are matched to commands through word patterns and synonyms (e.g., "details", "harder" and "longer" all mean more detail); about thirty English phrases and twenty-eight Hindi or Hinglish forms are recognised. The application listens after reading the home and examples screens, after the microphone button or a two-finger upward swipe, and when it asks whether the text was clear. An optional always-listening mode is intended for headphone users, since through the loudspeaker the microphone can pick up the application's own speech.

### 4.7 Touch Layer and Screen-Reader Support

In touch mode, a transparent layer recognises taps, double taps, holds, swipes and two-finger gestures. Each screen defines its controls in reading order; swiping highlights a control and announces its name, description and position, and a double tap activates it. While text is being read, a double tap with nothing selected pauses or resumes, swiping up or down changes the level, and any touch first pauses reading. The Phase 1 mapping, in which swipes moved directly between sentences, is kept as a separate gesture mode with keyboard shortcuts in the accessibility version; the default changed because trials showed that users first needed a way to discover the controls.

In screen-reader mode the touch layer is disabled. Controls have accessible names, levels and languages are radio groups, word explanations and help are dialogs that move focus to their close control, announcements use ARIA live regions [1], and Hindi sentences carry a language attribute so supported screen readers switch to a Hindi voice.

### 4.8 Hindi Support and Visual Design

Selecting Hindi changes the simplified text, the application's spoken prompts and the recognition language. Commands include "*ruko*" (pause), "*aage*" (next), "*phir se*" (repeat), "*aur aasaan*" (simpler) and "*mool*" (original), plus common Hinglish forms (Appendix A). The rupee sign is spoken as "*rupaye*". The Original level and on-screen button labels remain in English.

The visual design serves low-vision users and sighted helpers: a light, high-contrast theme, body text of about 18 px, buttons at least 52 px high and larger home-screen actions. The current sentence is highlighted, difficult words are underlined, and an optional control shows the original wording under each simplified sentence.

## 5 Prototype Walkthrough and Iterations

### 5.1 Walkthrough

The opening screen offers the three modes; one tap starts the voice-guided touch mode. The home screen offers three actions—paste text, photograph a document or hear an example—and a language choice. Four fictional example texts are built in, each with prepared levels, a glossary and two to four comprehension questions: a library note used for practice (three parts), a medicine leaflet for "Lisinor 10 mg" (eight parts), a bank KYC letter (seven parts) and an electricity bill (six parts). During reading, the current sentence is highlighted and the listener can switch level, show the original, open a word explanation or the help screen. Screenshots of these screens, including Hindi mode and the safety fallback, are in Appendix B.

Code repository: *[https://github.com/hemanitekwani/SimpliSpeak]*

### 5.2 Iterative Refinement

The application was developed over several build cycles. In each cycle the team carried out the core tasks on a real phone, recorded where the interaction broke down and revised the design. This formative testing does not replace a study with blind and low-vision users, but it removed several barriers before such a study. The main findings fall into four themes.

**Discoverability.** In the first build an invisible gesture layer blocked the visible buttons, so the single audio mode was split into the three modes of Section 3.4. Controls were hard to find when nothing was being read, which led to swipe exploration that announces each control's name and position, a yellow focus ring

for low-vision users, a voice-guided home screen with three large actions, and an on-demand help sheet.

**Voice and touch control.** Commands such as "pause" were missed while the application was speaking, so the vocabulary was expanded and push-to-talk now pauses reading first. A double tap with nothing selected gave confusing instructions and now simply pauses or resumes. The close control of word explanations was hard to find, so explanations became a dialog that moves focus to it and speaks how to continue.

**Language and safety.** Hindi output was initially paired with English-only commands; Hindi recognition, vocabulary and prompts were added. Simplified Hindi sometimes dropped dates or numbers, which motivated the fuller-level fallback with a Hindi warning.

**Reliability and presentation.** Reaching the free AI quota during testing led to the model chain, answer cache and offline rules. A dark theme was disliked, so a light high-contrast theme is used throughout.

# 6 Evaluation

## 6.1 Technical Verification

Before involving participants, we checked that the implemented features work as intended, in two ways. First, automated tests ran the application in a phone emulator in Chrome. They replaced the browser's speech output and recognition with scripted versions that recorded everything the application said and fed it spoken replies, so the spoken interaction itself could be checked. Second, the team used the complete English and Hindi interaction on a real phone. Table 2 shows which features each method covered. All automated checks passed, except three checks in an early run of the touch-mode walkthrough; these were mistakes in the test script, not in the application. Only the optional always-listening mode has not yet been tried on a real phone. None of these checks involved blind or low-vision participants, so they show that the features work, not that they help users.

**Table 2: Features verified by automated tests in a phone emulator and by the team on a real phone.**

| Feature | Emulator | Real phone |
|---|---|---|
| Touch-mode navigation | Passed | Tested |
| Reading flow and microphone | Passed | Tested |
| Word explanations | Passed | Tested |
| English voice commands | Passed | Tested |
| Paste and photo input | – | Tested |
| Hindi voice commands | Passed | Tested |
| Hindi simplification and reading | Passed | Tested |
| Hindi safety fallback | Passed | Tested |

## 6.2 Indicative Response Times

Table 3 lists individual measurements taken on a home connection with a free-tier API key; they are not benchmarks. While models were busy, one Hindi simplification took about 94 s and fell back to English, and one photo took about 9 s, showing the dependence on external model availability.

**Table 3: Indicative response times during development.**

| Operation | Time |
|---|---|
| Simplifying a 55-word, three-sentence text (default / minimal thinking) | 13.7 s / 4.8 s |
| Hindi simplification via backup model | 5–8 s |
| Word explanation | ≈3 s |
| Reading text from a photo | ≈1.8 s |
| Cached example or repeated text | <1 s |

## 6.3 Planned User Study

No user study has been conducted. The study mode implements the planned counterbalanced within-subjects procedure for blind and low-vision participants, who will use their preferred assistive technology; participants are not assumed to share the same level of screen-reader experience. Condition A reads the text as written with only next, back and repeat, similar to a conventional screen reader; Condition B uses the full SimpliSpeak interaction. After a practice task with the library note, each participant hears the medicine leaflet and the bank letter, one per condition, with text and condition order counterbalanced over four groups (participant number modulo four). A final task uses the electricity bill in Condition B, entered by photographing a printed copy, so that the two input methods—pasting copied text and photographing paper—can be compared.

After each task the application asks spoken comprehension questions and the NASA Task Load Index (NASA-TLX), a standard questionnaire on perceived mental effort [8]. At the end it presents the System Usability Scale (SUS), a standard ten-item usability questionnaire [5], followed by ratings of trust and control, preference questions and six interview questions. Every action—gestures, voice commands, level changes, word requests and timings—is logged to a JSON file per participant and analysed by `analysis/analyze.py`. Table 4 lists the planned measures. Comprehension questions will target critical facts such as instructions, dates, amounts and warnings rather than recall of wording.

**Table 4: Planned evaluation questions and measures.**

| Question | Measure |
|---|---|
| Can users complete tasks without looking at the screen? | Completion rate, time on task, navigation errors and requests for help. |
| Does simplification improve understanding? | Correct answers to questions on key facts. |
| How much effort does it take? | NASA-TLX workload questionnaire after each task. |
| Is it easy to use and trustworthy? | SUS usability questionnaire; interview questions on confidence and control. |
| Which simplification levels do users choose? | How often and when users change level, per document. |
| Which input method do users prefer? | Time, errors and preference for pasting text versus photographing a page. |

## 7 Ethics and Risk Assessment

**Meaning distortion.** The most serious risk is that simplification changes or omits information: an altered dose, contraindication, amount or deadline could cause harm or a poor decision. The guard (Section 4.4) covers supported numbers and negations only, so human review may be needed for high-stakes use, and SimpliSpeak is not a substitute for professional medical, legal or financial advice.

**Privacy.** User text and photos are sent to Google's Gemini API. Answers are cached on the server in .cache/ai_answers.json, which may contain personal text and must be deleted after demonstrations and studies. The API key stays on the server; the phone stores only preferences such as speed and language; there are no accounts. All study texts are fictional. Before use with real documents, users must be told that content is sent to an external service, and an on-device model, a retention limit and a user-facing deletion control should be considered.

**Accessibility and inclusion.** Implementing accessible labels, live regions, focus management and three interaction modes does not show that they work for everyone. Gesture conflicts, discoverability, speech recognition and screen-reader compatibility must be evaluated with users of differing assistive-technology experience.

**Consent and burden.** Studies will use accessible information and consent materials, voluntary participation with the right to pause or withdraw, fictional documents, and no questions about diagnoses or unnecessary personal information.

## 8 Limitations and Future Work

(1) **No user study.** Claims about comprehension, workload, usability, trust and independence are untested; the personas are illustrative, and the gesture mappings have not been validated against users' screen-reader habits.
(2) **Limited semantic checking.** The guard does not detect changed names, added advice or shifts in meaning; for instance, a backup model turning a possibility into a prohibition would pass.
(3) **Cloud dependence.** Models may be busy or rate-limited, backup models can give weaker Hindi, and sending documents to a cloud service raises privacy concerns.
(4) **Device-dependent speech.** Voices vary by phone, Hindi requires an installed Hindi voice, and recognition needs internet access and works best with headphones.
(5) **Prototype deployment.** The server runs on a development machine rather than a hosted service, so the prototype is not yet publicly available.
(6) **Language and input.** Button labels and the Original level remain in English; PDF, Word, email and share-sheet input are unavailable; texts are limited to 3,000 words.
(7) **Study mode** has not been re-tested since the latest changes to the speech module.

Future work will run the planned study with blind and low-vision participants; add entailment-based checks for changed or added content; explore on-device or self-hosted models; add PDF import and share-sheet support; provide Hindi labels and further Indian languages; and deploy the server as a hosted service or build a native application.

## 9 Conclusion

SimpliSpeak has progressed from a low-fidelity concept to a working application. A listener can bring in text by pasting, typing or photographing it; hear it sentence by sentence in English or Hindi; change the simplification level without losing their place; ask for word meanings; and return to the exact original, using voice, touch, large buttons or their own screen reader. A protected-facts guard falls back to fuller wording when checked facts are lost. Automated emulator tests and real-phone team testing show that the interaction works as designed, but not that it helps its intended users. The planned study with blind and low-vision participants is the essential next step: SimpliSpeak should be judged by whether it helps listeners understand information while keeping control and confidence in what they hear.

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## A Command Reference

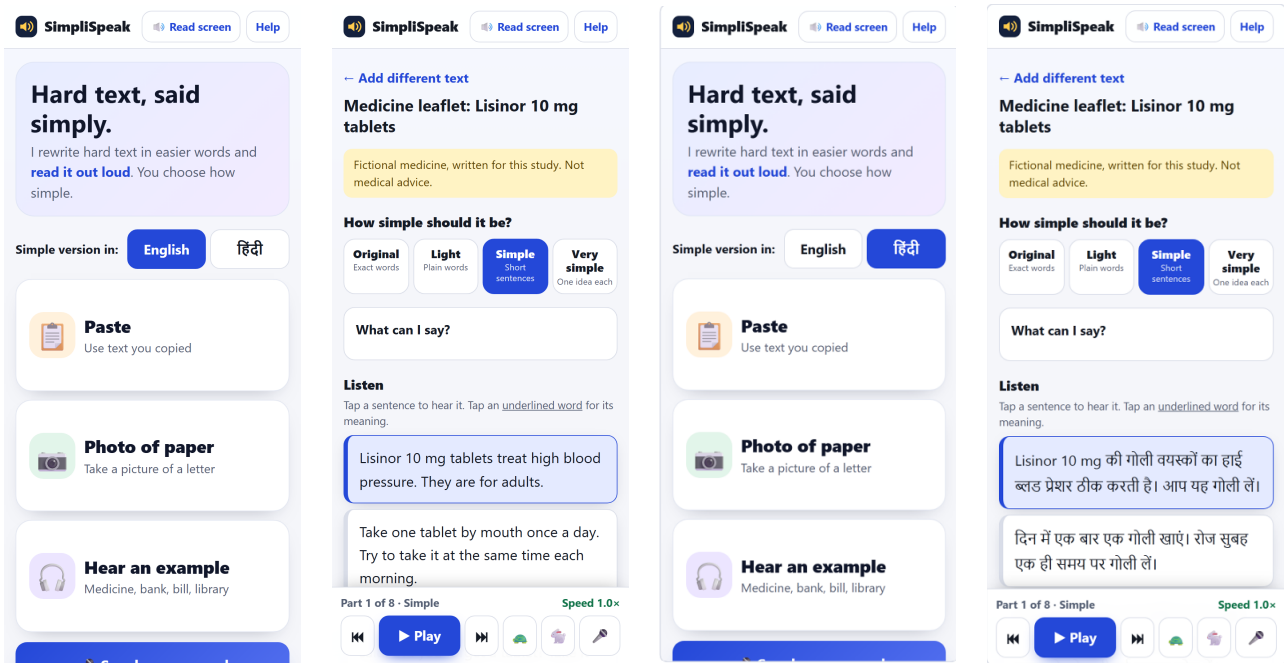
Table 5 lists how each main action is performed in each modality. Synonyms and common Hinglish forms of the voice commands are also accepted.

**Table 5: Main commands by modality.**

| Action | Voice (English) | Voice (Hindi) | Touch mode | Button |
|---|---|---|---|---|
| Pause / play | pause / play | *ruko* / *chalao* | Double tap with nothing selected, or two-finger tap | Play / pause |
| Next / back | next / back | *aage* / *peechhe* | Swipe to Next or Previous, then double tap | Next / previous |
| Repeat | repeat | *phir se* | No direct gesture | Tap the sentence |
| Simpler / more detail | simpler / more detail | *aur aasaan* / *vistaar se* | Swipe up / down while reading | Level buttons |
| Original | original | *mool* | Swipe to Original, then double tap | Original |
| Word meaning | what does X mean | X *ka matlab kya hai* | Two-finger swipe up, then speak | Tap underlined word |
| Hear the screen | read screen | – | Hold | Read screen |
| Speed | faster / slower | *tez* / *dheere* | Swipe to speed control, then double tap | Speed buttons |

## B Application Screenshots

Figure 2 shows the main English and Hindi screens.




(a) English home (b) English reading (c) Hindi home (d) Hindi reading

**Figure 2: SimpliSpeak on a phone: (a) home screen with input and language options; (b) reading a sentence at the Simple level; (c) home screen with Hindi selected; (d) a medicine leaflet simplified into Hindi.**